

RAUDTEEALASED RAKENDUSED.
ELEKTROMAGNETILINE ÜHILDUVUS. OSA 3-2: VEEREM.
APARATUUR

Railway applications - Electromagnetic compatibility --
Part 3-2: Rolling stock - Apparatus

EESTI STANDARDI EESSÕNA

NATIONAL FOREWORD

See Eesti standard EVS-EN 50121-3-2:2015 sisaldab Euroopa standardi EN 50121-3-2:2015 ingliskeelset teksti.	This Estonian standard EVS-EN 50121-3-2:2015 consists of the English text of the European standard EN 50121-3-2:2015.
Standard on jõustunud sellekohase teate avaldamisega EVS Teatajas	This standard has been endorsed with a notification published in the official bulletin of the Estonian Centre for Standardisation.
Euroopa standardimisorganisatsioonid on teinud Euroopa standardi rahvuslikele liikmetele kättesaadavaks 27.03.2015.	Date of Availability of the European standard is 27.03.2015.
Standard on kättesaadav Eesti Standardikeskusest.	The standard is available from the Estonian Centre for Standardisation.

Tagasisidet standardi sisu kohta on võimalik edastada, kasutades EVS-i veebilehel asuvat tagasiside vormi või saates e-kirja meiliaadressile standardiosakond@evs.ee.

ICS 29.020, 29.080, 45.060.01

Standardite reprodutseerimise ja levitamise õigus kuulub Eesti Standardikeskusele

Andmete paljundamine, taastekitamine, kopeerimine, salvestamine elektroonsesse süsteemi või edastamine ükskõik millises vormis või millisel teel ilma Eesti Standardikeskuse kirjaliku loata on keelatud.

Kui Teil on küsimusi standardite autorikaitse kohta, võtke palun ühendust Eesti Standardikeskusega:

Aru 10, 10317 Tallinn, Eesti; koduleht www.evs.ee; telefon 605 5050; e-post info@evs.ee

The right to reproduce and distribute standards belongs to the Estonian Centre for Standardisation

No part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying, without a written permission from the Estonian Centre for Standardisation.

If you have any questions about copyright, please contact Estonian Centre for Standardisation:

Aru 10, 10317 Tallinn, Estonia; homepage www.evs.ee; phone +372 605 5050; e-mail info@evs.ee

English Version

Railway applications - Electromagnetic compatibility - Part 3-2: Rolling stock - Apparatus

Applications ferroviaires - Compatibilité électromagnétique -
Partie 3-2: Matériel roulant - Appareils

Bahnwendungen - Elektromagnetische Verträglichkeit -
Teil 3-2: Bahnfahrzeuge - Geräte

This European Standard was approved by CENELEC on 2015-01-05. CENELEC members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration.

Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the CEN-CENELEC Management Centre or to any CENELEC member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CENELEC member into its own language and notified to the CEN-CENELEC Management Centre has the same status as the official versions.

CENELEC members are the national electrotechnical committees of Austria, Belgium, Bulgaria, Croatia, Cyprus, the Czech Republic, Denmark, Estonia, Finland, Former Yugoslav Republic of Macedonia, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, the Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey and the United Kingdom.



European Committee for Electrotechnical Standardization
Comité Européen de Normalisation Electrotechnique
Europäisches Komitee für Elektrotechnische Normung

CEN-CENELEC Management Centre: Avenue Marnix 17, B-1000 Brussels

Contents

Foreword.....	3
1 Scope	4
2 Normative references	4
3 Terms, definitions and abbreviations.....	5
4 Performance criteria	6
5 Conditions during testing	6
6 Applicability.....	7
7 Emission tests and limits	7
8 Immunity tests and limits	12
Annex A (informative) Examples of apparatus and ports	17
Annex B (informative) Conducted disturbances generated by power converters.....	23
Annex ZZ (informative) Coverage of Essential Requirements of EU Directives	24

Foreword

This document (EN 50121-3-2:2015) has been prepared by CLC/TC 9X: "Electrical and electronic applications for railways".

The following dates are fixed:

- latest date by which this document has to be implemented at national level by publication of an identical national standard or by endorsement (dop) 2016-01-05
- latest date by which the national standards conflicting with this document have to be withdrawn (dow) 2018-01-05

This document supersedes EN 50121-3-2:2006.

EN 50121-3-2:2015 includes the following significant technical changes with respect to EN 50121-3-2:2006:

- clarification of scope (Clause 1);
- set dated normative references (Clause 2);
- new definition of ports and clarification in Tables 1 to 6;
- emission requirement extended in the frequency range 1 GHz to 6 GHz following EN 61000-6-4;
- immunity requirement extended in the frequency range 5,1 GHz to 6 GHz;
- revision of Annex B.

This European Standard is to be read in conjunction with EN 50121-1.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CENELEC [and/or CEN] shall not be held responsible for identifying any or all such patent rights.

This document has been prepared under a mandate given to CENELEC by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For the relationship with EU Directive(s) see informative Annex ZZ, which is an integral part of this document.

This standard forms Part 3-2 of the European Standard series EN 50121, published under the general title "Railway applications - Electromagnetic compatibility". The series consists of:

- Part 1: General
- Part 2: Emission of the whole railway system to the outside world
- Part 3-1: Rolling stock - Train and complete vehicle
- Part 3-2: Rolling stock - Apparatus
- Part 4: Emission and immunity of the signalling and telecommunications apparatus
- Part 5: Emission and immunity of fixed power supply installations and apparatus

1 Scope

This European Standard applies to emission and immunity aspects of EMC for electrical and electronic apparatus intended for use on railway rolling stock. EN 50121-3-2 applies for the integration of apparatus on rolling stock.

The frequency range considered is from DC to 400 GHz. No measurements need to be performed at frequencies where no requirement is specified.

The application of tests shall depend on the particular apparatus, its configuration, its ports, its technology and its operating conditions.

This standard takes into account the internal environment of the railway rolling stock and the external environment of the railway, and interference to the apparatus from equipment such as hand-held radio-transmitters.

If a port is intended to transmit or receive for the purpose of radio communication (intentional radiators, e.g. transponder systems), then the radiated emission requirement in this standard is not intended to be applicable to the intentional transmission from a radio-transmitter as defined by the ITU.

Immunity limits do not apply in the exclusion bands as defined in the corresponding EMC related standard for radio equipment.

This standard does not apply to transient emissions when starting or stopping the apparatus.

The objective of this standard is to define limits and test methods for electromagnetic emissions and immunity test requirements in relation to conducted and radiated disturbances.

These limits and tests represent essential electromagnetic compatibility requirements.

Emission requirements have been selected so as to ensure that disturbances generated by the apparatus operated normally on railway rolling stock do not exceed a level which could prevent other apparatus from operating as intended. The emission limits given in this standard take precedence over emission requirements for individual apparatus on board the rolling stock given in other standards.

Likewise, the immunity requirements have been selected so as to ensure an adequate level of immunity for rolling stock apparatus.

The levels do not however cover all cases which may occur with an extremely low probability of occurrence in any location. Specific requirements which deviate from this standard shall be specified.

Test requirements are specified for each port considered.

These specific provisions are to be used in conjunction with the general provisions in EN 50121-1.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 50121-1:2015, *Railway applications - Electromagnetic compatibility - Part 1: General*

EN 50121-3-1:2015, *Railway applications - Electromagnetic compatibility - Part 3-1: Rolling stock - Train and complete vehicle*

EN 50155:2007, *Railway applications - Electronic equipment used on rolling stock*

EN 55016-1-1:2010, *Specification for radio disturbance and immunity measuring apparatus and methods - Part 1-1: Radio disturbance and immunity measuring apparatus - Measuring apparatus (CISPR 16-1-1:2010)*

EN 55016-1-4:2010, *Specification for radio disturbance and immunity measuring apparatus and methods - Part 1-4: Radio disturbance and immunity measuring apparatus - Antennas and test sites for radiated disturbance measurements (CISPR 16-1-4:2010)*

EN 55016-2-1:2009, *Specification for radio disturbance and immunity measuring apparatus and methods - Part 2-1: Methods of measurement of disturbances and immunity - Conducted disturbance measurements (CISPR 16-2-1:2008)*

EN 55016-2-3:2010, *Specification for radio disturbance and immunity measuring apparatus and methods - Part 2-3: Methods of measurement of disturbances and immunity - Radiated disturbance measurements (CISPR 16-2-3:2010)*

EN 55022:2010, *Information technology equipment - Radio disturbance characteristics - Limits and methods of measurement (CISPR 22:2008, mod.)*

EN 61000-4-2:2009, *Electromagnetic compatibility (EMC) - Part 4-2: Testing and measurement techniques - Electrostatic discharge immunity test (IEC 61000-4-2:2008)*

EN 61000-4-3:2006, *Electromagnetic compatibility (EMC) - Part 4-3: Testing and measurement techniques - Radiated, radio-frequency, electromagnetic field immunity test (IEC 61000-4-3:2006)*

EN 61000-4-4:2004, *Electromagnetic compatibility (EMC) - Part 4-4: Testing and measurement techniques - Electrical fast transient/burst immunity test (IEC 61000-4-4:2004)*

EN 61000-4-5:2006, *Electromagnetic compatibility (EMC) - Part 4-5: Testing and measurement techniques - Surge immunity test (IEC 61000-4-5:2005)*

EN 61000-4-6:2009, *Electromagnetic compatibility (EMC) - Part 4-6: Testing and measurement techniques - Immunity to conducted disturbances, induced by radio-frequency fields (IEC 61000-4-6:2008)*

EN 61000-4-30:2009, *Electromagnetic compatibility (EMC) - Part 4-30: Testing and measurement techniques - Power quality measurement methods (IEC 61000-4-30:2008)*

EN 61000-6-4:2007, *Electromagnetic compatibility (EMC) - Part 6-4: Generic standards - Emission standard for industrial environments (IEC 61000-6-4:2006)*

3 Terms, definitions and abbreviations

3.1 Terms and definitions

For the purposes of this document, the following terms and definitions apply

3.1.1

rolling stock apparatus

a finished product with an intrinsic function intended for implementation into the rolling stock installation

3.1.2

port

the particular interface of the specified apparatus with the external environment e.g. auxiliary AC or DC power port, I/O (input/output) port

3.1.3

enclosure port

the physical boundary of the apparatus through which electromagnetic fields may radiate or impinge

The main categories of ports for rolling stock apparatus are presented in Figure 1.

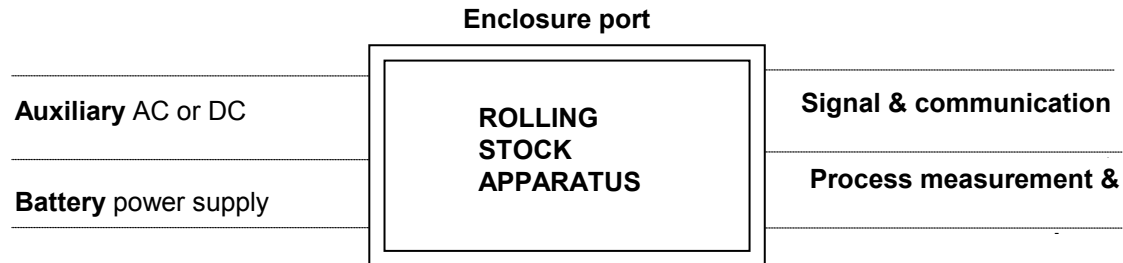


Figure 1 – Main categories of ports

Typical examples of rolling stock apparatus with their ports are listed in Annex A.

Traction power ports are not covered in EN 50121-3-2:2015, see Annex B.

3.2 Abbreviations

a.c.	Alternating current
AM	Amplitude modulation
CISPR	Comité international spécial des perturbations radioélectriques
d.c.	Direct current
EMC	Electromagnetic compatibility
EUT	Equipment under test
I/O	Input / Output
ITU	International Telegraph Union
OATS	Open area test site
PC	Personal computer
SAC	Semi-anechoic chamber
THD	Total harmonic distortion
TV	Television

4 Performance criteria

The variety and the diversity of the apparatus within the scope of this standard make it difficult to define precise criteria for the evaluation of the immunity test results.

A functional description and a definition of performance criteria, during or as a consequence of the EMC testing, shall be provided by the manufacturer and noted in the test report, based on the criteria A, B, C defined in EN 50121-1:2015.

5 Conditions during testing

It is not always possible to test every function of the apparatus. The tests shall be made at a typical operating mode considered by the manufacturer to produce the largest emission or maximum

susceptibility to noise as appropriate in the frequency band being investigated consistent with normal applications. The conditions during testing shall be defined in a test plan (see basic standard of EN 61000-4-x series).

If the apparatus is part of a system, or can be connected to auxiliary apparatus, then the apparatus shall be tested while connected to the minimum configuration of auxiliary apparatus necessary to exercise the ports in accordance e.g. with EN 55022:2010, Clause 8.

The configuration and mode of operation shall be specified in the test plan and the actual conditions, during the tests, shall be precisely noted in the test report.

If the apparatus has a large number of similar ports or ports with many similar connections, then a sufficient number shall be selected to simulate actual operating conditions and to ensure that all the different types of termination are covered (e.g. 20 % of the ports or at least four ports).

The tests shall be carried out within the specified operating range for the apparatus and at its nominal supply voltage, unless otherwise indicated.

6 Applicability

The measurements in this standard shall be made on the relevant ports of the apparatus.

It may be determined from consideration of the electrical characteristics, the connection and the usage of a particular apparatus that some of the tests are not applicable (e.g. radiated immunity of induction motors, transformers). In such cases, the decision not to test has to be recorded in the test plan and test report.

If not otherwise specified, the EMC tests shall be type tests.

7 Emission tests and limits

The emission tests and limits for apparatus covered by this standard are given on a port by port basis.

Measurements shall be performed in well-defined and reproducible conditions for each type of disturbance.

The description of the test, the test methods and the test set-up are given in Basic Standards which are referred to in Tables 1 to 3.

The contents of these Basic Standards are not repeated here, however modifications or additional information needed for the practical application of the tests are given in this standard.

NOTE The reference to "Basic Standard" is intended to be limited to those parts of the standard that give the description of the test, the test methods and the test set-up.

Table 1 – Emission – Auxiliary AC or DC power ports (input and output)

	Port	Test specification		Basic Standard	Test set-up	Applicability note	Remarks
1.1	Auxiliary supply sinusoidal AC or DC (port 9 on Figures A.1, A.2 and A.4)	150 kHz to 500 kHz	99 dB μ V peak	EN 55016-2-1	EN 55016-2-1	See a, b and c	For the time being there are no limits for shore supply mode. Therefore the limits given in this table are valid. Other limits may apply if connected e.g. to the public low voltage power supply and should be specified by the train operator.
1.2	AC power outlet port for public use	500 kHz to 30 MHz	93 dB μ V peak				
		50 Hz to 2 kHz	THD < 8 % (THD: harmonic distortion)	EN 61000-4-30			230 V AC power outlet ports for public use shall offer a power quality, which is sufficient for the use of intended equipment like PC and mobile telephone chargers. The harmonic distortion in differential mode shall be limited by a sine-filter to < 8 %.
a Wherever applicable the method defined by EN 55016-2-1 is to be used. At present the existing method of measuring conducted emissions (EN 55016-2-1) has limitations in terms of voltage and current rating of coupling networks. In addition the method of measuring voltage has safety implications for testing high power systems. Limiting conducted emissions from apparatus connected to external cable systems will prevent excessive radiated emissions. b This requirement refers to the industrial limit values but considering they have been defined to protect radio and TV sets and as the objective is not the same here, the applicable limit for railway applications have been relaxed by 20 dB to be more representative of potential problems. c This requirement is not applicable to power ports which are connected to other dedicated, compatible ports.							

Table 2 – Emission – Battery power supply (Input and output)

	Port	Test specification		Basic standard	Test set-up	Applicability note	Remarks
2.1	Battery power supply (port 10 on Figures A.1-A.5)	150 kHz to 500 kHz	99 dBµV peak	EN 55016-2-1	EN 55016-2-1	See d	
		500 kHz to 30 MHz	93 dBµV peak				
d This requirement refers to the industrial limit values but considering they have been defined to protect radio and TV sets and as the objective is not the same here, the applicable limit for railway applications have been relaxed by 20 dB to be more representative of potential problems.							

Table 3 - Emission - Enclosure port

	Port	Frequency range	Limits	Basic standard	Applicability note	Remarks
3.1	Enclosure	30 MHz to 230 MHz	40 dB μ V/m quasi-peak at 10m	The measurement instrumentation shall be as defined in Clause 4 of EN 55016-1-1:2010. The measuring antennas shall be as defined in 4.4 of EN 55016-1-4:2010. The measuring site shall be as described in Clause 5 of EN 55016-1-4:2010.	See e, f, g, h and k	May be measured at 30 m distance using the limits decreased by 10 dB. As stated in EN 55016-2-3 the antenna height shall be varied between 1 m to 4 m. Additional guidance on the test method can be found in EN 55016-2-3:2010, 7.3 and Clause 8.
3.2	Enclosure	1 GHz to 3 GHz	76 dB μ V/m peak at 3m 56 dB μ V/m avg. at 3m	The measurement instrumentation shall be as defined in Clauses 5 and 6 of EN 55016-1-1:2010. The measuring antennas shall be as defined in 4.5 of EN 55016-1-4:2010. The measuring site shall be as described in Clause 8 of EN 55016-1-4:2010.	See f, g, i, j and k	May be measured at greater distances with the limits decreased by 20 dB/decade (relative to distance) For SAC and OATS facilities absorber may be required to achieve free space conditions as defined in EN 55016-1-4.
		3 GHz to 6 GHz	80 dB μ V/m peak at 3m 60 dB μ V/m avg. at 3m	The measurement method shall be as specified in 7.3 of EN 55016-2-3:2010.		
e	Measurement distance is 10 m. A measurement distance of 3 m may be used with the limit increased by 10 dB.					
f	Traction converters and auxiliary converters over 50 kVA cannot be tested individually but when the vehicle is tested as a whole in accordance with EN 50121-3-1:2015.					
g	For apparatus containing devices operating at frequencies less than 9 kHz, measurements only need to be performed up to 230 MHz.					
h	The apparatus is deemed to comply with the enclosure port requirement below 1 GHz if it meets the requirements defined in one or more of the tables of Clauses 1.1, 1.2 or 1.3 in EN 61000-6-4:2007.					
i	If the highest internal frequency of the EUT is less than 108 MHz, the measurement shall only be made up to 1 GHz.					

If the highest internal frequency of the EUT is between 108 MHz and 500 MHz, the measurement shall only be made up to 2 GHz.

If the highest internal frequency of the EUT is between 500 MHz and 1 GHz, the measurement shall only be made up to 5 GHz.

If the highest internal frequency of the EUT is above 1 GHz, the measurement shall be made up to 6 GHz.

Where the highest internal frequency is not known, tests shall be performed up to 6 GHz.

j The peak detector limits shall not be applied to disturbances produced by arcs or sparks that are high voltage breakdown events. Such disturbances arise when devices contain or control mechanical switches that control current in inductors, or when devices contain or control subsystems that create static electricity (such as paper handling devices). The average limits apply to disturbances from arcs or sparks, and both peak and average limits will apply to other disturbances from such devices.

k At transitional frequencies, the lower limit applies.

8 Immunity tests and limits

The immunity tests and limits for apparatus covered by this standard are given on a port by port basis.

To ensure the immunity of the complete vehicle, the limits shall be applicable to all relevant apparatus.

Tests shall be conducted in a well-defined and reproducible manner.

The tests shall be carried out as single tests in sequence. The sequence of testing is optional. The description of the test, the test generator, the test methods and the test set-up are given in Basic Standards which are referred to in Tables 4 to 6.

The contents of these “Basic Standards” are not repeated here, however modifications or additional information needed for the practical application of the tests are given in this standard.

When installing new apparatus in existing rolling stock or replacing apparatus special care should be taken to ensure that EMC is maintained e.g. in the EMC management plan.

Table 4 - Immunity - Battery referenced ports (except at the output of energy sources), auxiliary AC power input ports (rated voltage $\leq 400 V_{rms}$)

	Environmental phenomena	Test specification		Basic Standard	Test set-up	Applicability note	Remarks	Performance criteria
4.1	Radio-frequency common mode	0,15 MHz 80 MHz 10 V (rms) 80 % AM, 1 kHz	to	EN 61000-4-6	EN 61000-4-6	See I	The test level specified is the rms value of the unmodulated carrier	A
4.2	Fast transients	± 2 kV 5/50 ns 5 kHz	Peak T_r / T_h Repetition frequency	EN 61000-4-4	EN 61000-4-4	see m	Direct coupling	A
4.3	Surges	1,2 / 50 μ s ± 2 kV 42 Ω , 0,5 μ F ± 1 kV 42 Ω , 0,5 μ F	Open circuit test voltage, line to ground Open circuit test voltage, line to line	EN 61000-4-5	EN 61000-4-5	see m, n	All severity levels below the given severity level have to be tested with each 5 pulses and a test sequence not alternating but first one polarity, then the other polarity Test with maximum permanent supply voltage as defined in EN 50155	B
I The test level can also be defined as the equivalent current into a 150 Ω load. m The test applies to the power supply ports and the battery control input output ports, with direct coupling, positive and negative polarity. n The test applies to the power supply ports and the battery control input output ports. This test is intended to replicate the phenomenon known as direct coupling; hence an output impedance of 42 Ω (40 Ω and 2 Ω generator) and a coupling capacitance of 0,5 μF is specified. This value is accepted as a compromise as the impedance of battery referenced network inside a rolling stock can vary up to a few hundred ohms depending on their length.								

Table 5 - Immunity - Signal and communication, process measurement and control ports

	Environmental phenomena	Test specification		Basic Standard	Test set-up	Applicability note	Remarks	Performance criteria
5.1	Radio-frequency common mode	0,15 MHz 80 MHz 10 V (rms) 80 % AM, 1 kHz	to Unmodulated carrier	EN 61000-4-6	EN 61000-4-6	See p, q	The test level specified is the rms value of the unmodulated carrier	A
5.2	Fast transients	±2 kV 5/50 ns 5 kHz	Peak T_r / T_h Repetition frequency	EN 61000-4-4	EN 61000-4-4	See o, p	Capacitive clamp used	A
o Capacitive coupling, positive and negative polarity. Battery referenced control ports are covered by requirements in Table 4. p Applicable only to ports interfacing with cables whose total length according to the manufacturer's functional specification may exceed 3m. q The test level can also be defined as the equivalent current into a 150 Ω load.								

Table 6 - Immunity - Enclosure ports

	Environmental phenomena	Test specification		Basic Standard	Test set-up	Applicability note	Remarks	Performance criteria
6.1	Radio-frequency electromagnetic field. Amplitude modulated	80 MHz to 800 MHz 20 V/m (rms) 80 % AM, 1 kHz	Unmodulated carrier	EN 61000-4-3	EN 61000-4-3	See r and s	The test level specified is the rms value of the unmodulated carrier	A
6.2	Radio-frequency electromagnetic field, from digital communication devices	800 MHz to 1 000 MHz 20 V/m (rms) 80 % AM, 1 kHz 1 400 MHz to 2 000 MHz 10 V/m (rms) 80 % AM, 1 kHz 2 000 MHz to 2 700 MHz 5 V/m (rms) 80 % AM, 1 kHz 5 100 MHz to 6 000 MHz 3 V/m (rms) 80 % AM, 1 kHz	Unmodulated carrier	EN 61000-4-3	EN 61000-4-3	See s	The test level specified is the rms value of the unmodulated carrier	A

6.3	Electrostatic discharge	±6 kV ±8 kV	Contact discharge Air discharge	EN 61000-4-2	EN 61000-4-2	See t	B
r This limit applies to equipment mounted in the passenger compartments, drivers cab or external to the rolling stock (roof, underframe). For equipment mounted in all other areas a severity level of 10 V/m may be used s For large apparatus (e.g. traction drives, auxiliary converters) it is often not practical to perform the immunity test to radiated electromagnetic fields on the complete unit. In such cases the manufacturer should test susceptible sub-systems (e.g. control electronics). The test report should justify the selection or not of sub-systems and any assumptions made (e.g. reduction of field due to case shielding). t Only applicable to equipment accessible to passengers and operational staff (not maintenance).							

Annex A (informative)

Examples of apparatus and ports

The purpose of this annex is to provide examples of the different types of rolling stock apparatus together with their ports. Examples of apparatus which may be placed on the market as a single commercial unit are given in Table A.1. However these items of apparatus may also form a sub-system in a larger apparatus (e.g. control electronics in an auxiliary converter). In this case, the requirements of the standard apply only to the apparatus which is placed on the market. A port is defined in the standard as the interface of an apparatus with the external environment. The matrix in the table indicates whether the particular apparatus is relevant to emission, immunity or neither. This guidance is offered for the benefit of users of this standard, but it is not intended to be definitive. It is for the user of the standard to make the necessary technical judgements in determining whether or not a test is applicable.

The drawings on the following figures clarify the most essential ports. They show examples of different arrangements.

Figure A.1 applies for an AC fed locomotive with AC traction drive and psophometric filter on the line side.

Figure A.2 shows another AC/AC system with power factor correction filter on the converter side and with DC or three-phase auxiliary and train power supply.

Figure A.3 shows a more conventional system with AC input and DC traction motors fed by phase control converter.

Figure A.4 is a DC fed system with AC traction drive.

Figure A.5 shows some additional ports of converter and control electronics.

Of course, many other different system arrangements are possible.

Table A.1 – Typical examples of apparatus

Apparatus	Test requirements
Traction converter	Emission and immunity
Main circuit breaker	No test requirements
Traction transformers	No test requirements
Traction motor	No test requirements
Auxiliary motor (e.g. induction motor for fans)	No test requirements
DC auxiliary supply (battery)	Emission and immunity
Signalling and communication equipment	Emission and immunity
Electronic man-machine interface	Emission and immunity
Environmental conditioning equipment	Emission and immunity
Passenger information equipment	Emission and immunity
Door control	Emission and immunity
Auxiliary equipment for train operation	Emission and immunity
Auxiliary equipment for passenger services	Emission and immunity
Train management systems	Emission and immunity
Electronic power supply	Emission and immunity
Braking control system	Emission and immunity

In Tables 1 to 6, tests are specified for application to a specific port (interface of an apparatus). Table A.2 lists some more typical descriptions used for these ports and the type of apparatus which may have such a port. Examples of these ports are given in the accompanying figures except for ports n° 11, 13 and 14.

Table A.2 – Typical port descriptions

Port No on figures	Typical port name	Typical apparatus
	Traction AC power ports	
1	Pantograph line terminal	Main circuit breaker
3	High voltage connection (before filter)	Filter
4	Connection filter-transformer, HV side	Filter
5	Train power line single phase	Auxiliary converter
6	Connection transformer-converter	Propulsion converter
7	Traction motor cables	Traction motors
8	Auxiliary feed windings of transformer	DC auxiliary supply
	Traction DC power ports	
2	DC conductor input	Main circuit breaker
3	High voltage connection (before filter)	Filter
6	Connection filter-converter	Propulsion converter
7	Traction motor cables	Traction motors
	Auxiliary AC ports	
9	Auxiliary AC supply	Environmental conditioning equipment
	Auxiliary DC ports	
9	Auxiliary DC supply	
	Battery referenced ports	
10	Battery power supply	Electronic power supply
11	Train control bus (conventional battery voltage)	Train management system
19	Relay logic input/output	Electronic control system
	Signal and communication ports	
12	Databus within vehicle	Electronic control system
13	Databus within train	Train management system
14	Passenger entertainment network	Passenger entertainment equipment
15	Firing control line	Electronic control system
17, 18	Sensor/transducer signal (digital or analog)	Electronic control system
20	Communication interface (maintenance)	Electronic control system
	Process measurement and control ports	
16	Internal electronic supply	Electronic control system
18	Sensor/transducer signal (analog)	Electronic control system
	Enclosure port	
21	Equipment enclosure	All apparatus
	Earth port	
22	Earth connection	All apparatus

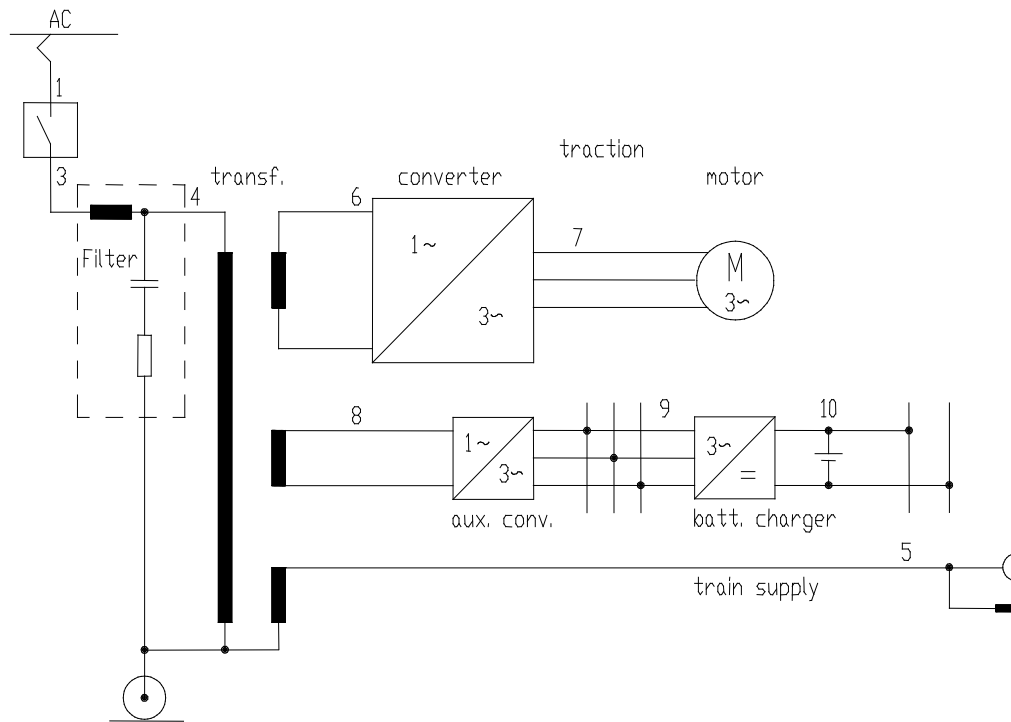


Figure A.1 – AC fed loco with AC traction drive and psophometric filter on the line side

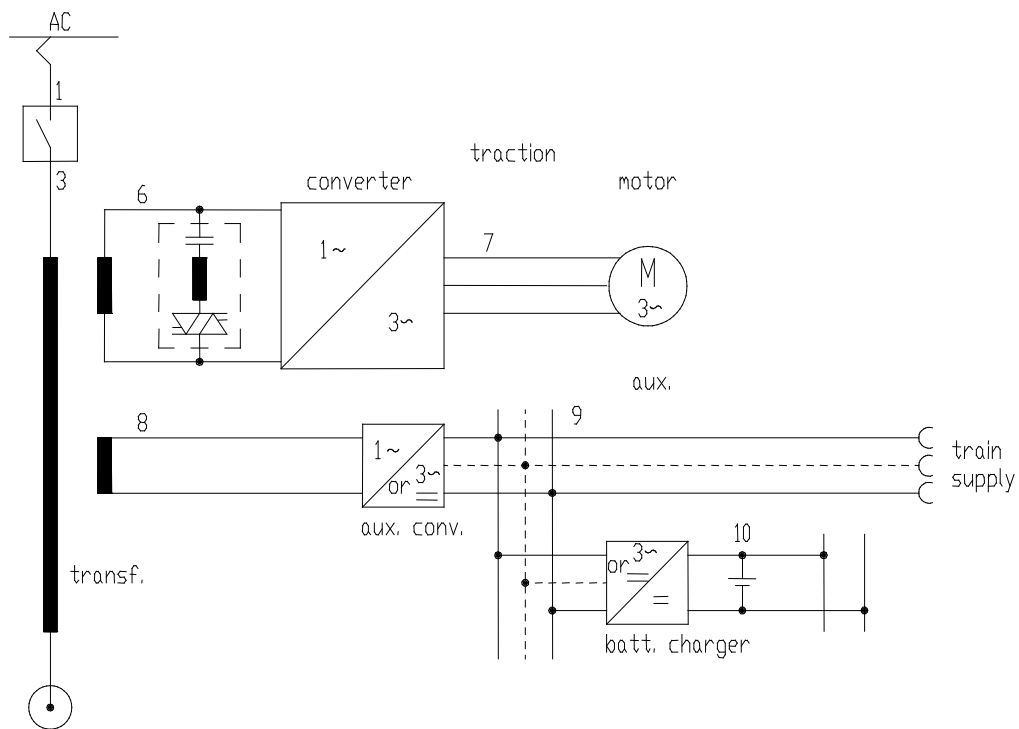


Figure A.2 – AC/AC system with power factor correction filter on the converter side and with DC or three-phase auxiliary and train power supply

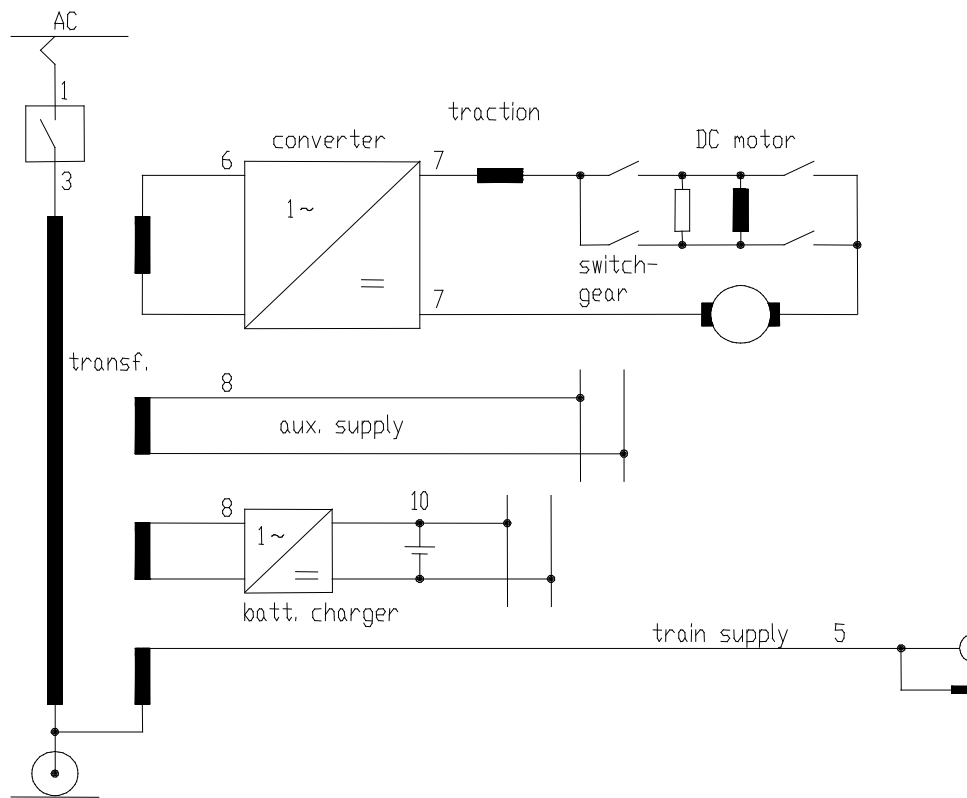


Figure A.3 – Conventional system with AC input and DC traction motors fed by phase control converter

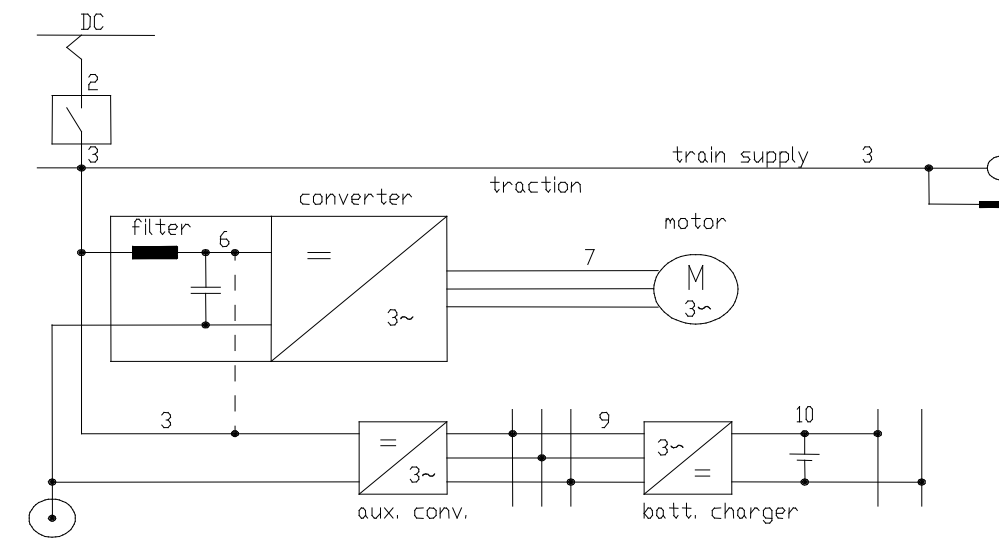


Figure A.4 – DC fed system with AC traction drive

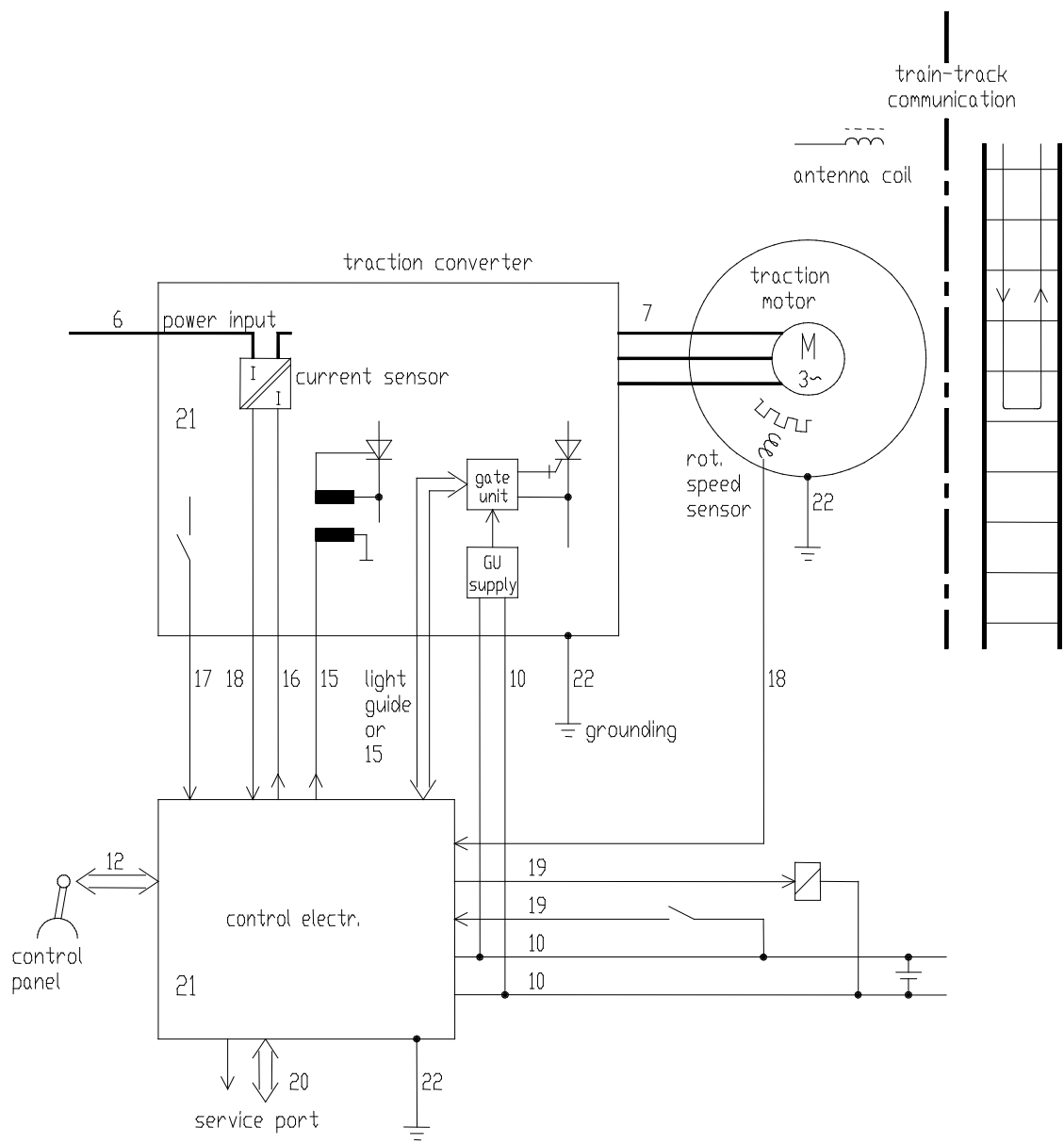


Figure A.5 – Additional ports of converter and control electronics

Annex B (informative)

Conducted disturbances generated by power converters

No limits apply for laboratory tests. Emission requirements for this type of equipment are tested during the emission test of the complete rolling stock.

Table B.1 reminds the requirements for AC and DC ports.

Table B.1 - Emission requirements for AC and DC power ports

	Port	Test specification		Remarks
B.1	AC (port 3 on Figures A.1, A.2, A.3) DC (port 3 on Figure A.4)	Signalling and telecommunication frequencies	see national requirements, contractual requirements, TSI and/ or EN 50238 series	Note 1
		150 kHz to 30 MHz	EN 50121–3-1:2015, 6.2.3	Note 2
NOTE 1: Homologation tests will be performed on the complete train.				
NOTE 2: The requirement in EN 50121–3-1:2015 is expressed at the train level as a radiated emission limit.				

Some preliminary tests during the design phase could provide helpful information about the emission expected.

Figure B.1 shows a possible test set-up.

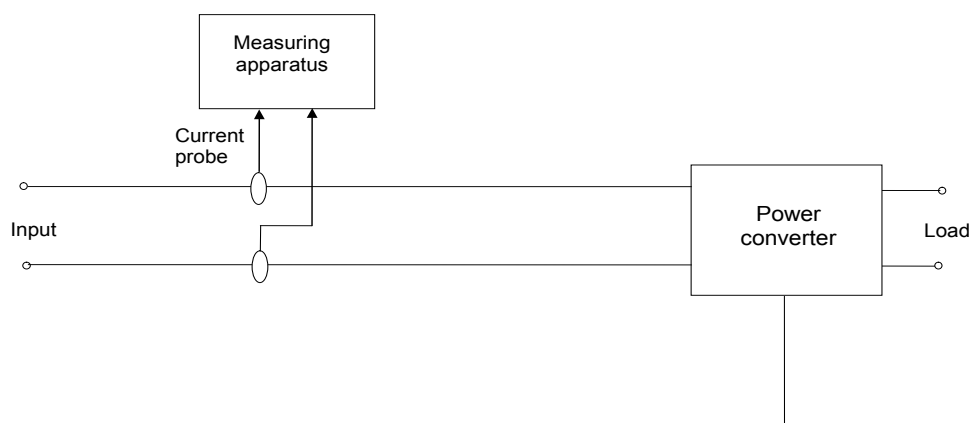


Figure B.1 – Test set-up

Annex ZZ (informative)

Coverage of Essential Requirements of EU Directives

This European Standard has been prepared under a mandate given to CENELEC by the European Commission and the European Free Trade Association and within its scope the standard covers all relevant essential requirements as given in Annex I, Article 1 of the EC Directive 2004/108/EC.

Compliance with this standard provides one means of conformity with the specified essential requirements of the Directive concerned.

WARNING: Other requirements and other EU Directives may be applicable to the products falling within the scope of this standard.

[Blank page]

standard@evs.ee
www.evs.ee